UN ESCAP REGTECH AI HACKATHON 2026 · RDTII 2.1 · PILLARS 6 & 7

ClauseChain

A reproducible proof for every exported legal row.

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ONE SLIDE

Executive Summary

WHAT	An autonomous legal-evidence engine that turns official statute portals, gazette PDFs and treaty texts into RDTII 2.1 indicator evidence — every row carrying a byte-exact verbatim snippet, pinpoint citation, archived source and a verification receipt.
PROBLEM	RDTII evidence collection is manual, slow and fragile: scattered portals, anti-bot walls, scanned gazettes, bilingual statutes, dead links, and citation errors that are expensive for lawyers to catch.
CORE TECH	Format-based extractors (HTML / PDF / EPUB / OCR) → statute-structure RuleUnit graph (SQLite / Neo4j) → hybrid retrieval (exact-phrase + BM25 + dense) → LLM mapping with rubric-as-code → 9 deterministic gates → adversarial refuter → named human review → deterministic replay.
WHY P6 & P7 FIT	The rubric is encoded as data (YAML), not code: indicator polarity, 0.5 tiers, the 7.5 court-order test, and 6.5 treaty evidence all flow through one generic pipeline — the same design that scales to the 8 finals economies.

WHY THIS IS HARD

Problem Statement

1

Fragmented sources

Evidence lives across statute portals, regulator PDFs, gazettes and treaty registers — each with its own markup, structure and access rules.

2

Hostile acquisition

CloudFront/Akamai anti-bot walls, geo-blocks and TLS fingerprinting; links in official inventories die (we found dead links in the provided inventory itself).

3

Hard documents

Scanned gazettes need OCR; consolidated acts exceed 600 pages; Malaysia publishes bilingual text ("Seksyen 12A"); treaties use Article grammar.

4

Legal precision

A quote that stops mid-clause, a repealed act cited as current, or a wrong pinpoint costs real credibility — lawyers must verify in seconds, not hours.

5

Polarity traps

7.1/7.2 score the ABSENCE of frameworks; 7.5 turns on a court-order test; 6.4-vs-6.1 hinges on condition-vs-ban — keyword matching gets these wrong.

6

Proving absence

"No provision found" is a claim about the whole corpus — it must be backed by a search-coverage manifest, not silence.

WHAT CLAUSECHAIN IS DESIGNED TO ACCOMPLISH

Objective

Autonomous discovery	Crawl official portals and the ESCAP legal inventory; acquire, archive and hash every source; discover provision-level NEW evidence beyond the sample kit — the 20-point lever.
Faithful extraction	Parse statutes into their real structure (Part → section → subsection), never fixed-size chunks; OCR scanned gazettes with CER < 5%; keep original-language verbatim text.
Verified mapping	Map provisions to P6-I1…P7-I5 with rubric-as-code + LLM reasoning, then force every row through 9 deterministic gates before it may exist in the output.
Human-authoritative output	Role-separated named review (citation vs mapping), append-only decision receipts, and a deterministic replay that regenerates the final CSV/JSON from approvals alone.

EVIDENCE FLOW, SOURCE TO SUBMISSION

System Architecture

SOURCES	SSO · lom.agc · legislation.gov.au · gazettes · MITI/ESG treaty registers
ACQUIRE	Seeds pipeline · politeness + Playwright anti-bot fallback · archive + SHA-256 · reconciliation report
EXTRACT	Format-based: anchored HTML · statute PDF · EPUB oracle · OCR VM · grammar profiles (treaty / Malay)
GRAPH	RuleUnit store (SQLite default, Neo4j opt-in) · FTS5 + embeddings · cross-references
MAP & GATE	Hybrid retrieval → LLM screen/map (nano→mini escalation) → gates G1–G9 → NEW/KNOWN diff
VERIFY & SHIP	Adversarial refuter · named human review · zone-3 noise audit · replay → CSV/JSON

Technology Solution & Innovative Features

STACK & LICENSING

Engine — Python 3.12 (PSF), httpx/pydantic/numpy (BSD/MIT), pymupdf (AGPL, unmodified library use)

Graph — SQLite + FTS5 (public domain) default · Neo4j Community (GPLv3) optional via GRAPH_BACKEND

Models — profile-swappable via models.yaml: cloud GPT nano/mini tiers (operator key) or local Ollama; BGE-M3 embeddings (MIT); local OCR-VLM

App — Django + DRF (BSD), Next.js/React (MIT), PostgreSQL (PostgreSQL licence)

Ops — systemd + nginx on plain Linux; no proprietary services required; Path A runs fully offline from a clean clone

WHAT MAKES IT DIFFERENT

- Proof chain, not just answers — every row: archived bytes + SHA-256 + span-level citation proof + gate results
- Rubric-as-code — indicator polarity, 0.5 tiers, exclusions and treaty scoping live in YAML data, never in code
- Honest failure — dead links become ACQUISITION_UNRESOLVED facts that BLOCK absence conclusions (network failure is never legal absence)
- Processing fingerprints — source sha + extractor version + grammar profile gate every cache reuse; parser changes force re-extraction
- Gold-audit tripwire — we detected planted errors in the provided Malaysia gold data and filed corrections with receipts

CHUNKING = STATUTE STRUCTURE, NEVER FIXED WINDOWS

Backend Logic I — Discover, Extract, Structure

Discovery	data/seeds.json (ESCAP inventory + deep-research deltas) drives acquisition. Each build reconciles the manifest: cached-ok is never refetched, new rows are fetched, failures are recorded — and declared expectations (e.g. "Art. 14.11 must parse") fail the build closed if the wrong document came back.
Extraction	Extractors are organized by FILE FORMAT, not country: anchored-HTML (SSO print view, incl. subsidiary legislation), statute-PDF with named grammar profiles (Commonwealth "s. 12(1)", treaty "Art. 14.11", Malay "Seksyen 12A"), EPUB structure oracle aligned to the authorised PDF, OCR for scanned gazettes.
Chunking strategy	The unit of retrieval and citation is the RuleUnit: one provision at subsection depth (id, article path, verbatim text, parent-section context, page/anchor location, modality components). No sliding windows — a citation can always name its exact clause.
Graph	Instrument → Section → Provision nodes with CROSS_REFERENCES / AMENDS edges in a swappable GraphStore (SQLite judged path; Neo4j demo). Every provision carries legal status evidence, eligibility, source-artifact hash and processing fingerprint.

HYBRID RETRIEVAL · RUBRIC-AS-CODE MAPPING · 9 GATES

Backend Logic II — Retrieve, Map, Verify

Hybrid retrieval	Three legs, unioned broad-recall (never top-k): deterministic exact-phrase/defined-term scan, BM25 full-text (FTS5), dense cosine (cached embeddings). Master-known anchors are INJECTED into the candidate set so gold recall never depends on ranking. Source-type allowlists scope evidence classes per indicator (P6-I5 = treaties only) before any cap; every truncation is recorded.
Mapping	LLM screen (bulk tier) → mapping (high tier) with the indicator brief generated from YAML: legal test, polarity rule (absence-scored 7.1/7.2/6.5), exclusions, official scoring criteria. Low-confidence and ban-vs-condition ambiguity escalate to a stronger model; every rationale is stored.
Gates G1–G9	Deterministic, fail-closed, run on the FINAL exported text: G1 snippet is byte-exact in the archived source · G2 location · G3 official-domain authority · G4 currentness/status · G5 whole-rule · G6 rationale support · G7 indicator fit · G8 dangling references · G9 span-closure review flag.
Edge cases	Ambiguity is surfaced, not hidden: adversarial refuter attacks every NEW row; absence rows ship a search-coverage manifest and are blocked while any configured acquisition is unresolved; zone-3 scores get a multi-persona noise audit with Krippendorff's α and a master-gold tripwire.

MEASURED, LOGGED, REPRODUCIBLE

Evaluation Metrics & Performance

53,969

provisions in the judged corpus (SG 10,494 · MY 3,688 · AU 39,787)

9

deterministic gates on every row — snippet exactness is enforced by construction, not sampled

74

provision-level NEW findings discovered beyond the sample kit in the final sweep

148

automated tests green, incl. builder-level integration and champion-contract suites

HOW WE MEASURE

- Citation fidelity — G1 verifies each exported snippet byte-exact against the archived official copy; alignment proofs bind PDF page spans. Zero-tolerance: a row that fails any gate cannot be exported.
- Master-gold recall — every master-known anchor is tracked per run; misses are adjudicated into REAL_MISS / GOLD_WRONG / CORRECT_ABSTENTION with receipts (we filed corrections to the provided gold where its own citations were wrong).
- OCR quality — scanned-gazette path measured by character error rate (target < 5%), with per-page confidence and citation-token disagreement flags carried into review.
- Cost & speed — logs/cost_report.json meters every call; one economy-pillar run ≈ 7–15 min on the accuracy profile; embeddings cached, unchanged documents restamped, LLM spend only on changed evidence.

WHAT THE JUDGES WILL SEE

Demo & Preview

1 · One command	python run.py --economy Singapore --pillar 6 — clean clone, offline Path A, no keys. The same engine the web console triggers.
2 · Live web console	Admin-gated Runs screen starts a real sweep; Review screen walks findings with source context, gate results and refuter verdicts; decisions are written with named, role-separated reviewers.
3 · Scanned-PDF path	The ≤10-min recording processes a scanned gazette end-to-end: OCR → RuleUnits → mapped finding with page-anchored citation proof.
4 · Replay & receipts	submission_replay.py regenerates the final CSV/JSON purely from approvals; every bundle is content-hashed so the artifact is reproducible on the judges' machine.

THE BUILD STORY · CHAPTER ONE

Six walls.
Six break-throughs.

Everything on the next six slides actually happened during this build — and each fix left a receipt the engine still carries.

The portals fight back

THE WALL

Singapore sits behind CloudFront rate-limiting; Australia's treaty portal (dfat.gov.au) rejects the TLS handshake of anything that is not a real browser. Official inventories even carry dead links. A crawler that pretends this is fine produces silent gaps.

THE BREAKTHROUGH

Politeness + backoff first; a real-browser (Playwright) fallback second. And when a wall would not fall — DFAT stayed geo-blocked — the engine records ACQUISITION_UNRESOLVED instead of concluding "no treaty exists." Network failure is never legal absence.

RECEIPT Australia P6-I5 ships blocked-by-design, with five unresolved acquisitions listed in its coverage manifest — visible in the app's review screen.

The reference data lied

THE WALL

Malaysia’s provided reference data contains deliberately planted errors — and Malaysia counts double. One master row cites “clause 4.11” of the Personal Data Protection Act 709. We searched the official text; that clause does not exist.

THE BREAKTHROUGH

Every reference anchor is re-verified per run. Misses are adjudicated into REAL_MISS / GOLD_WRONG / CORRECT_ABSTENTION with named human decisions — so an error in the reference data becomes a documented correction, not a copied mistake.

RECEIPT Two GOLD_WRONG adjudications with reviewer receipts in data/review/recall_decisions.json; the audit trail is browsable in the Ledger.

Monster documents

THE WALL

A 666-page consolidated compilation. A telecommunications act published as two PDF volumes. Tables of contents that mimic real sections and poison naive parsers. Fixed-size chunking would shred provisions mid-sentence and cite none of them correctly.

THE BREAKTHROUGH

Statute-structure parsing: an EPUB structure oracle aligned span-exact to the authorised PDF, a monotonic section filter that kills list-item false positives, and RuleUnits at subsection depth — the unit of retrieval IS the unit of citation.

RECEIPT 39,787 Australian provisions parsed with page-anchored citations; Corporations Act 2001 alone contributes 14,078 aligned units.

The scanner's ghosts

THE WALL

Malaysian gazettes arrive as scans. OCR misreads statutory labels — a "12B" becomes "12b" or worse, and a wrong section label means a wrong citation in front of a lawyer. Bilingual acts print "Seksyen 12A" where the gold data says "s. 12A."

THE BREAKTHROUGH

The OCR router measures per-page confidence and carries it into review; a sequence-aware repair restores misread section labels from statutory order; a named Malay grammar profile parses Seksyen/Perkara headings while keeping Commonwealth citations.

RECEIPT Per-page OCR confidence + citation-token disagreement flags are visible on affected findings in the review screen.

The polarity trap

THE WALL

RDTII scores 7.1, 7.2 and 6.5 on the ABSENCE of frameworks. Our first treaty sweep proved how dangerous that is: retrieval surfaced 77 relevant treaty articles for P6-I5 — and the mapper rejected every single one for proving the “wrong” answer to the indicator question.

THE BREAKTHROUGH

Polarity is rubric DATA, not prompt folklore: absence-scored indicators carry an evidence rule that tells the mapper the row documents what EXISTS while the score captures absence. One YAML block fixed all 77; a regression test pins the behaviour forever.

RECEIPT The fix is one commit touching only configs/rdtii/pillar_6.yaml; the very next run produced CPTPP Art. 14.11 and DEPA Art. 4.3 as NEW rows.

The half-quoted law

THE WALL

An early export ended a quotation at "…may be issued with an or" — a truncated word. Another stopped at a list-introducing colon, quoting a rule while omitting every one of its conditions. A lawyer reading either would stop trusting everything else.

THE BREAKTHROUGH

Snippets are finalized source-exact and structurally closed BEFORE any gate runs: closure follows list children to a genuine sentence stop, legal abbreviations ("s.", "No.") never count as endings, and a passage whose closure cannot be proven cannot be exported at all.

RECEIPT The G9 structural-closure gate records closure codes on every row; unclosable spans fail closed and go to human review instead of the output file.

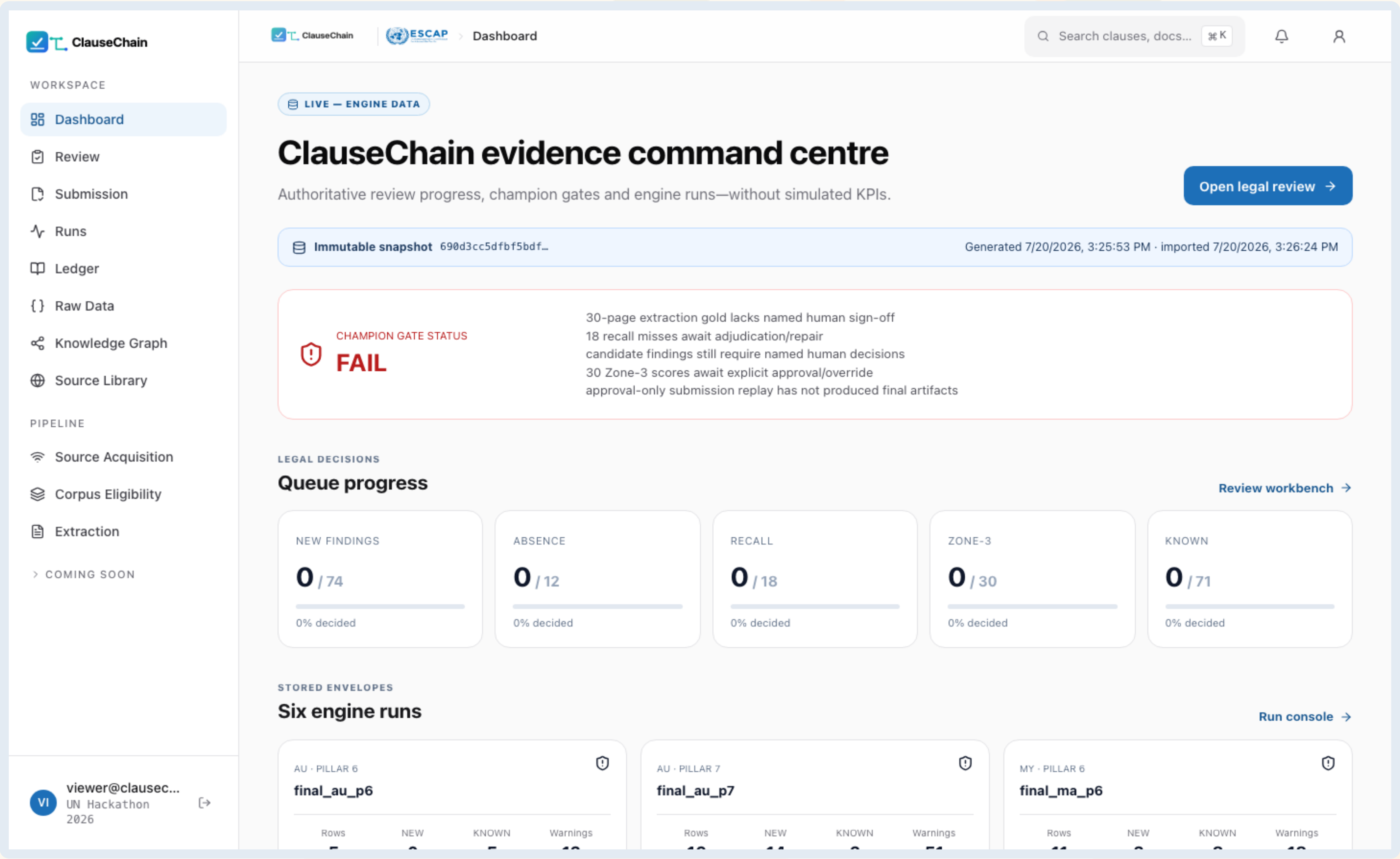
THE BUILD STORY · CHAPTER TWO

Proof, not promises.

A working system anyone can open today — read-only demo access is printed on the login page.
These are unedited screenshots.

PROOF · COMMAND CENTRE

The system refuses to flatter itself

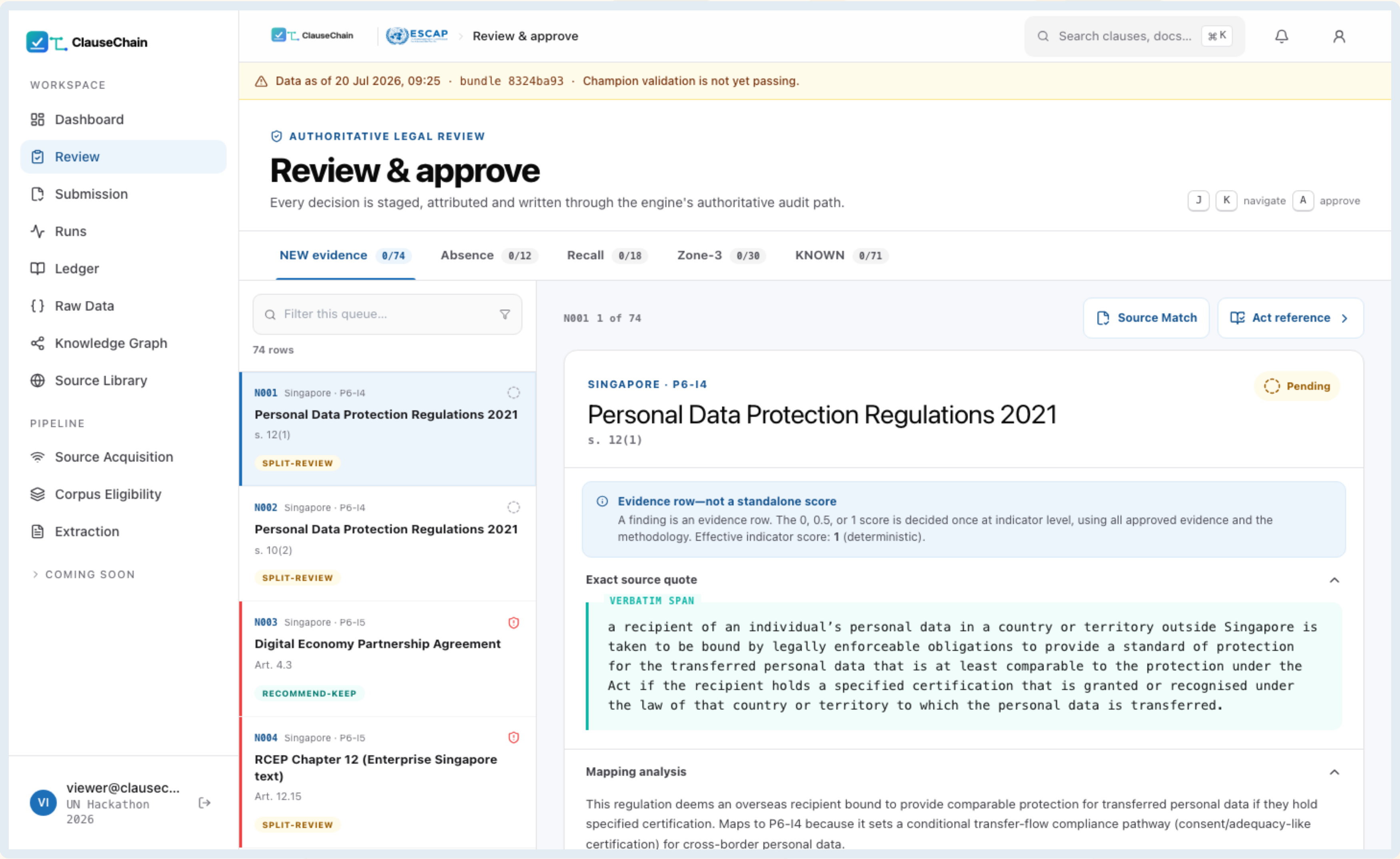


The dashboard's champion gate reads FAIL until every human obligation is met.

- Immutable snapshot hash on every screen — the app is a mirror of engine artifacts, never its own truth
- Live queue progress across NEW / absence / recall / zone-3
- No simulated KPIs anywhere — the banner says so

PROOF · REVIEW WORKBENCH

Where a lawyer meets the evidence

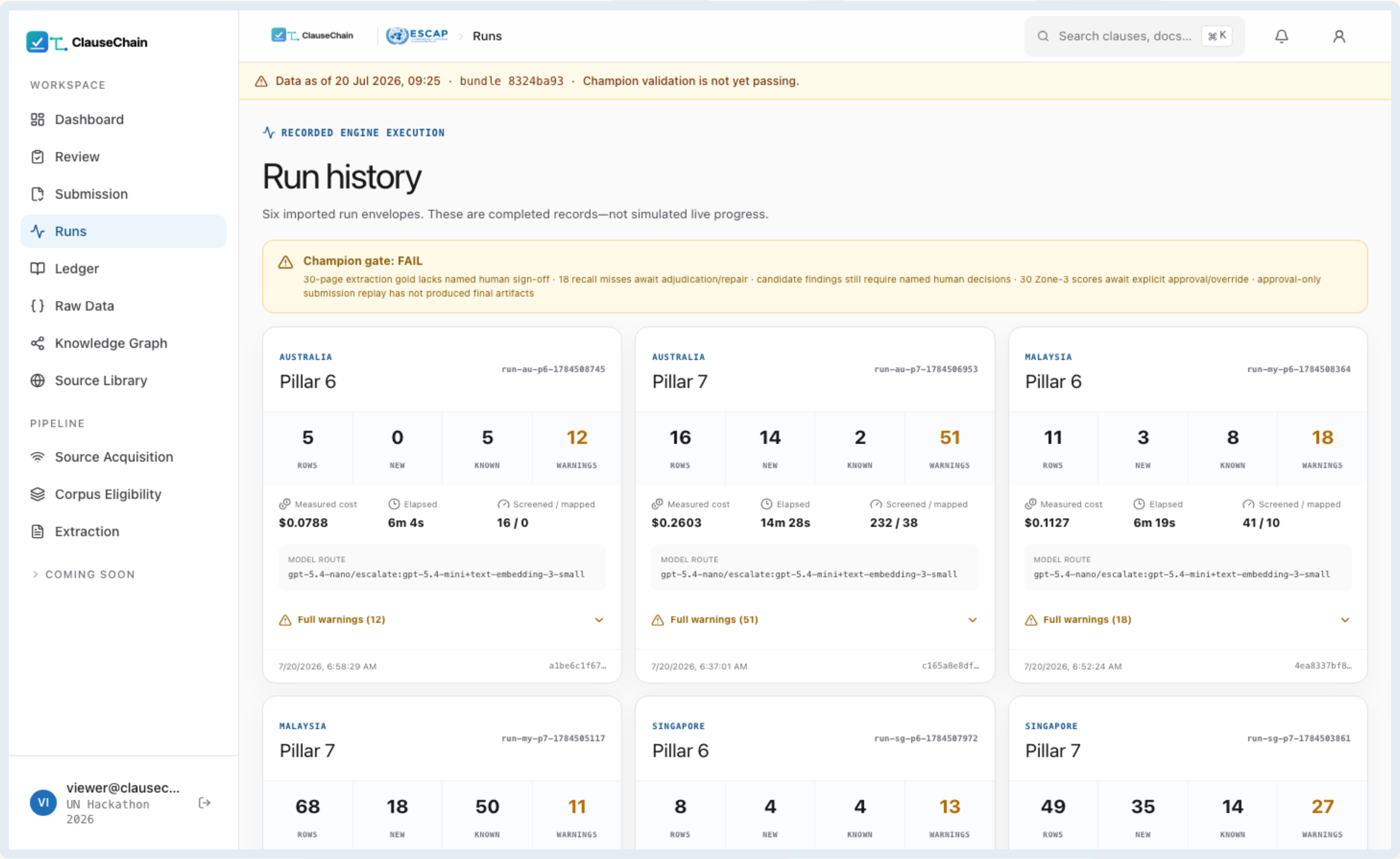


Every finding: exact source quote, mapping analysis, refuter verdict, staged sign-off.

- Verbatim span highlighted from the archived official copy
- Refuter chips (RECOMMEND-KEEP / SPLIT-REVIEW) triage attention
- Treaty rows N003/N004 — the first 6.5 evidence — sit in the same queue as everything else

PROOF · RUN HISTORY

Every run is an accountable record

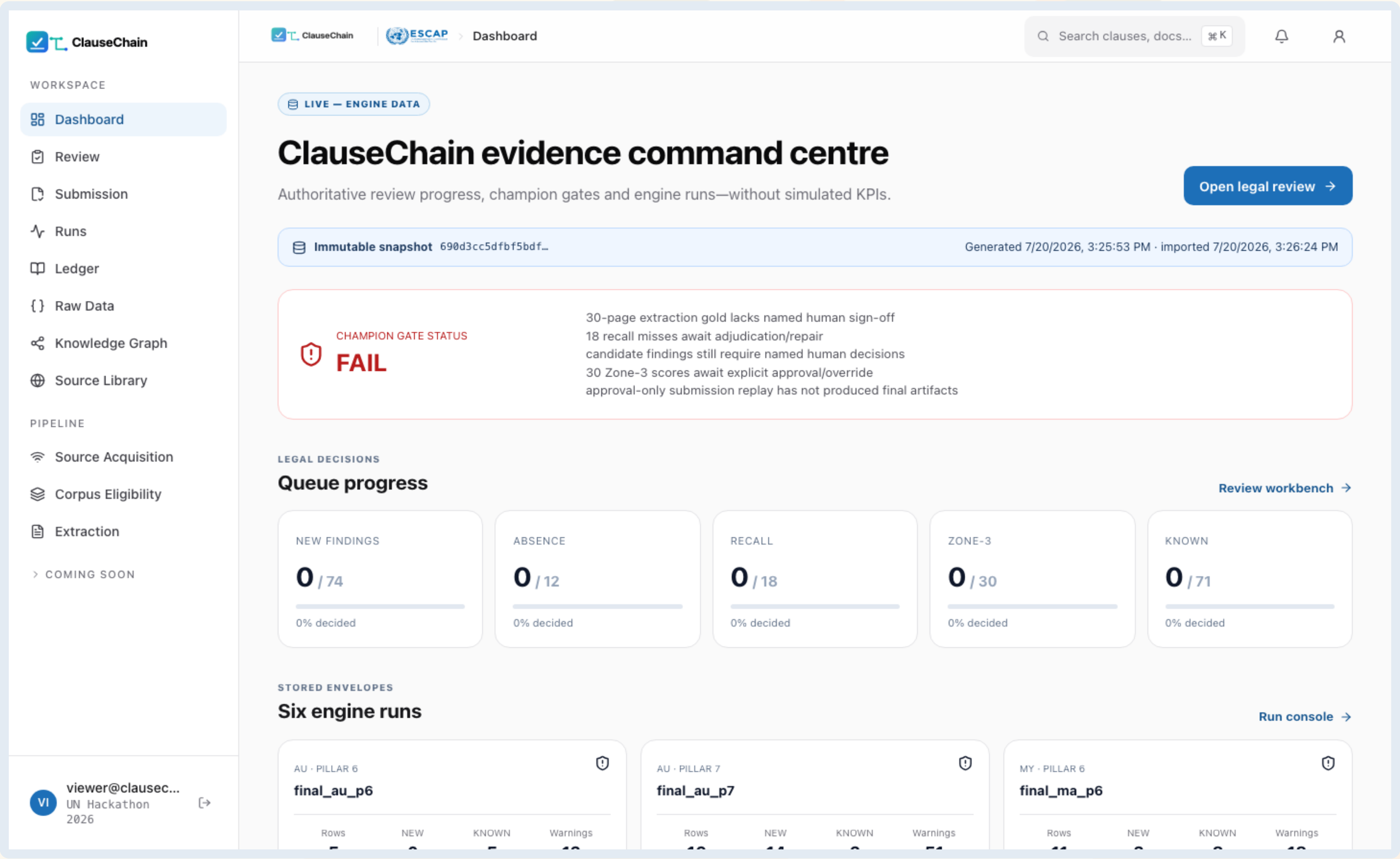


Six envelopes: measured cost, elapsed time, model route, warnings — per run.

- $0.08–$0.34 per economy-pillar, printed with the model route that produced it
- Warnings are first-class (“Full warnings (51)”), not buried in logs
- Content hash on each envelope ties the UI to the artifact

PROOF · PIPELINE

Watching the evidence arrive



Source acquisition with archive hashes, reconciliation and failure honesty.

- Each source: official URL, SHA-256, access date, status
- Dead links stay visible — they feed the audit, not the void
- The same screen a judge would use to trace any citation back

PROOF · SUBMISSION

The output file, with its receipts

ClauseChain

WORKSPACE

- Dashboard
- Review
- Submission
- Runs
- Ledger
- Raw Data
- Knowledge Graph
- Source Library

PIPELINE

- Source Acquisition
- Corpus Eligibility
- Extraction

› COMING SOON

VI viewer@clausec...
UN Hackathon 2026

ClauseChain · ESCAP › Submission

Search clauses, docs... ⌘K

Data as of 20 Jul 2026, 09:25 · bundle 8324ba93 · Champion validation is not yet passing.

CANDIDATE EVIDENCE + VERIFICATION

Submission explorer

Final artifacts are produced only by deterministic engine replay—not by this table.

No replayed final artifact is available
The table below contains candidates; pending or rejected rows are not silently exported.

Law, citation, indicator or quote... Search | All economies | All pillars | NEW + KNOWN | All review states

157 candidate rows

13 required template fields + mechanical verification

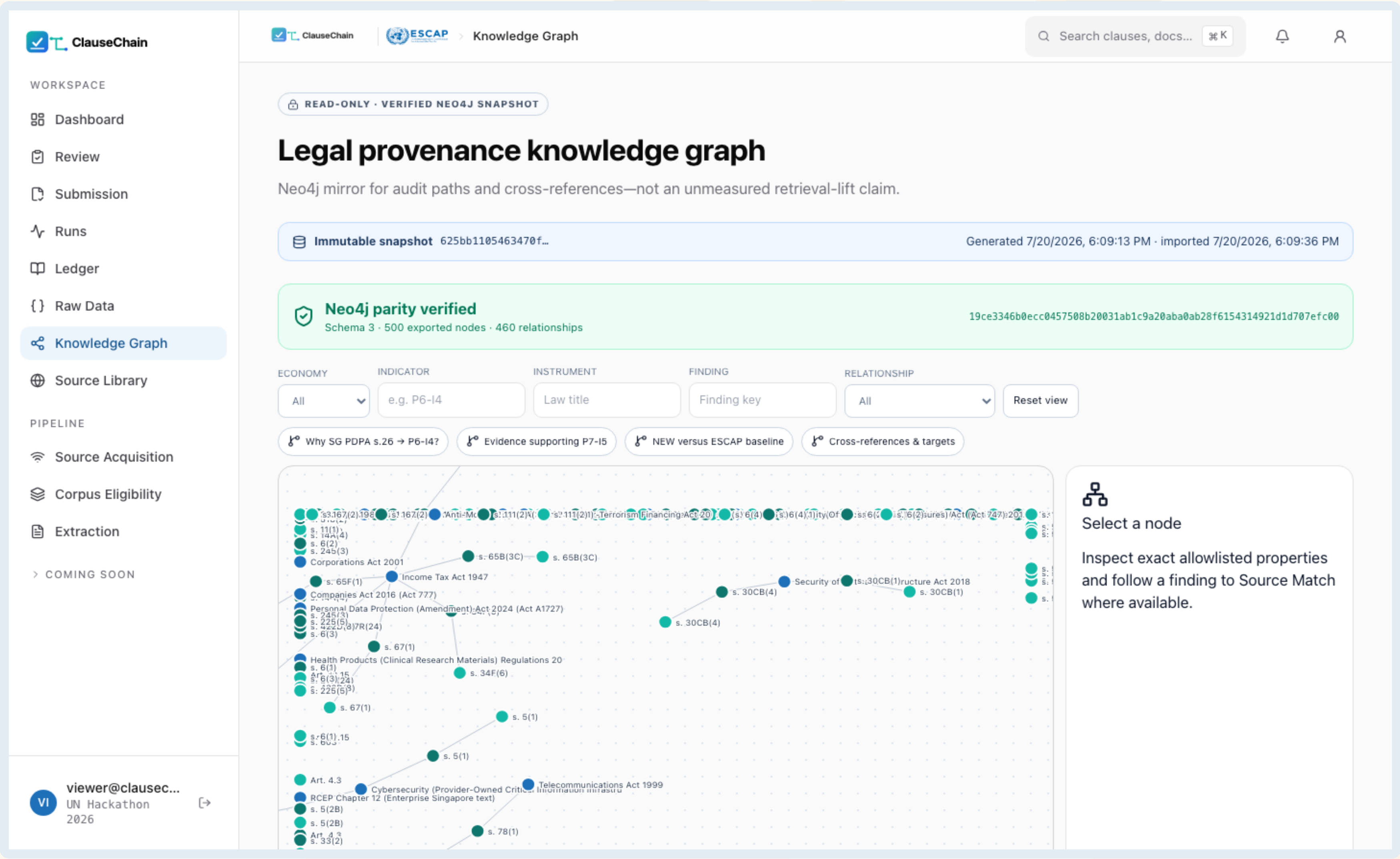
REVIEW	ECONOMY	LAW NAME	LAW NUMBER / REF	LAST AMENDED	INDICATOR ID	ARTICLE / SECTION	DISCOVERY TAG	LOCATION REFERENCE	VERBATIM SNIPPET
PENDING	Singapore	Personal Data Protection Act 2012	—	—	P6-I1	n/a	KNOWN	n/a	NO_EVIDENCE_FOUND_PENDING
PENDING	Singapore	Personal Data Protection Act 2012	—	—	P6-I2	n/a	KNOWN	n/a	NO_EVIDENCE_FOUND_PENDING
PENDING	Singapore	Personal Data Protection Act 2012	—	—	P6-I3	n/a	KNOWN	n/a	NO_EVIDENCE_FOUND_PENDING
PENDING	Singapore	Personal Data Protection Act 2012	No. 26 of 2012	—	P6-I4	s. 26(1)	KNOWN	#pr26-	An organisation must not transfer
PENDING	Singapore	Personal Data Protection Regulations 2021	—	—	P6-I4	s. 12(1)	NEW	#pr12-	a recipient of an individual's perso
PENDING	Singapore	Personal Data Protection Regulations 2021	—	—	P6-I4	s. 10(2)	NEW	#pr10-	A transferring organisation is take
PENDING	Singapore	Digital Economy Partnership Agreement	—	—	P6-I5	Art. 4.3	NEW	page 16	Each Party shall allow the cross-b
PENDING	Singapore	RCEP Chapter 12 (Enterprise Singapore text)	—	—	P6-I5	Art. 12.15	NEW	page 10	A Party shall not prevent cross-bo
PENDING	Singapore	Personal Data Protection Act 2012	No. 26 of 2012	—	P7-I1	s. 11(2)	KNOWN	#pr11-	An organisation is responsible for

The consolidated dataset beside the decision bundles that produced it.

- Template-exact columns, one consolidated CSV/JSON
- Content-hashed review bundles — append-only history
- Replay regenerates identical bytes from approvals alone

PROOF · KNOWLEDGE GRAPH

The law as a graph, not a pile of text



Instrument → Section → Provision with cross-references — browsable.

- CROSS_REFERENCES edges follow "subject to s. 26" chains
- SQLite default; Neo4j swap via one env var
- Every node carries legal status + eligibility + source hash

WHY THIS WINS ON TRUST

Innovation & Competitive Advantage

Treaty-evidenced 6.5

P6-I5 rows cite the actual binding commitments (CPTPP Art. 14.11, DEPA Art. 4.3, RCEP Art. 12.15) parsed from official state registers through the same generic pipeline — no hand-curated rows anywhere.

We audit the gold

The engine re-verifies the provided master data and caught its planted Malaysia errors — including citations to clauses that do not exist — filing corrections with evidence.

Absence is proven

Score-0 claims carry a search-coverage manifest; unresolved acquisitions (e.g. a geo-blocked treaty register) BLOCK the absence conclusion instead of faking it.

Uncertainty is modelled

Zone-3 scores ship a multi-persona noise audit (Krippendorff's α) and a master-gold tripwire — a band with receipts, never a bare number.

Human-authoritative by design

Citation reviewer ≠ mapping reviewer, enforced at write time; append-only receipts; the submission is replayable from decisions alone.

Generic to the core

Nothing in code names a statute or a row: targets live in seeds/YAML data. The rerun that added treaties, Malay grammar and subsidiary legislation changed configuration, not engine logic.

THE FINALS QUESTION, ANSWERED CONCRETELY

Adding an Economy Is a Data Change

configs/jurisdictions/th.yaml (all it takes)

```
jurisdiction: TH
source_format: pdf
languages: { primary: th, supported: [th, en] }
section_grammars: [thai]          # "มาตรา N" — named profile
official_sources:
  - { domain: ratchakitcha.soc.go.th,
      source_type: official_gazette }
citation_patterns:
  section: ["s. {n}", "มาตรา {n}"]
status_assertions:
  "PDPA B.E. 2562": { status: in_force, fact_url: "…" }

# + seed rows in data/seeds.json  (official URLs)
# + nothing else. No code.
```

WHY IT SCALES

- Extractors are format-based (HTML / PDF / EPUB / OCR), never country-based — a new economy reuses them untouched.
- Section grammars are named data profiles: treaty and Malay shipped this round; "Pasal", "Статья", "มาตรา" are one dictionary entry each.
- Rubric YAML is economy-independent — pillars are added the same way (P8 = one new YAML, zero engine changes).
- Cross-lingual retrieval (BGE-M3) + original-language verbatim + English column are already wired.
- Proof: this round, treaties + subsidiary legislation + Malay grammar were added mid-sprint as configuration — the engine diff was generic plumbing only.
- The government-verified Round-2 gold (CN·IN·ID·LA·MN·RU·TH) is already ingested as the 8-economy eval baseline.

FROM 3 ECONOMIES TO THE REGION

Scalability & Future Development

SCALE PATH

- New economy = data, not code: a jurisdiction YAML (portals, citation grammar, status assertions) + seed rows. Extractors are format-based; grammars are named profiles ("Pasal", "Статья", "มาตรา" next).
- Finals-ready gold: the government-verified Round-2 database (CN·IN·ID·LA·MN·RU·TH) is already ingested as the evaluation baseline for the 8-economy expansion.
- Multilingual by architecture: BGE-M3 cross-lingual retrieval, original-language verbatim + appended English column, per-language section grammars.
- Continuous monitoring: incremental fingerprints mean a nightly run re-extracts only changed documents — amendment tracking falls out of the cache design.

VALUE FOR ESCAP & STAKEHOLDERS

- ESCAP researchers — RDTII updates in hours instead of months, with every cell traceable to an archived official source.
- Governments — see exactly which provision produced each score; corrections become data changes with receipts.
- Businesses & trade lawyers — pinpoint, current, verifiable answers on cross-border data rules across jurisdictions.
- Expected outcomes — faster policy insight, fewer stale citations, lower compliance-research burden, and a reusable evidence commons for digital-trade regulation.

OUR CASE

Why ClauseChain can be a Good Solution

SUBSTANTIVE ACCURACY

- 74 provision-level NEW findings beyond the sample kit — the 20-point lever, systematically harvested
- First treaty-evidenced P6-I5 rows (CPTPP 14.11, DEPA 4.3, RCEP 12.15) from official registers
- Byte-exact citations a lawyer verifies in seconds; polarity traps encoded as rubric data
- We audited the gold itself and caught the planted Malaysia errors — with receipts

TECHNICAL RESILIENCE

- Live portals with anti-bot walls beaten (backoff + real-browser fallback); failures recorded, never faked
- Scanned gazettes: OCR with per-page confidence + statutory glyph repair; bilingual Malay grammar
- End-to-end with zero manual steps — and a web console that triggers the same engine
- Incremental fingerprints: nightly re-runs re-extract only changed law

ARCHITECTURE

- Swap LLM / OCR / graph by config — two shipped profiles, incl. a fully key-free local path
- Audit trail beyond the ask: verbatim snippet + archived source + SHA-256 + gate results + named review + replay
- Measured cost: US$1.16 for the entire 3-economy sweep (≈ $0.01 per document)
- Adding an economy or pillar = one YAML + seed rows; finals gold already ingested

Trust is the product: every row the judges check will check out.

REFERENCES

Sources, models & libraries

LEGAL DATA SOURCES

- RDTII 2.1 Guide, methodology sheet & Round-1/Round-2 databases (UN ESCAP)
- Singapore Statutes Online — sso.agc.gov.sg (Acts + subsidiary legislation print view)
- Laws of Malaysia (AGC), Federal Gazette, PDP Department, NACSA, MITI FTA portal
- Federal Register of Legislation — legislation.gov.au (authorised PDF + EPUB)
- Official treaty texts: CPTPP, RCEP, DEPA, SADEA, KSDPA (state registers)

MODELS & METHODS

- GPT nano/mini tiers via API (operator key) · local Ollama fallback profile
- BGE-M3 multilingual embeddings (MIT)
- Noise-audit uncertainty design after Qian Xiao (Maynooth), 12 Jun workshop
- Direct corpus interaction / broad-recall motivation: Li et al., arXiv:2605.05242
- GraphRAG legal schema after Chawuthai (KMITL), 12 Jun workshop

OPEN-SOURCE LIBRARIES

- Python · httpx · pydantic · numpy · pymupdf · pytest
- SQLite/FTS5 · Neo4j Community (optional)
- Django · DRF · Next.js · React · PostgreSQL
- Playwright (anti-bot fallback) · systemd/nginx deployment
- Full licence inventory in the repository README